

PROACTIVE PORTFOLIO MANAGEMENT IN MISSION-ORIENTED INNOVATION POLICY

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Abstract

Mission portfolio management is a critical ingredient in the success of mission-oriented innovation policies. This practice aims at ensuring that a consistent and tailor-made set of diverse projects and activities contribute together to the realisation of the mission objectives. It is essential for generating the systemic impacts that missions aim to deliver. However, many missions currently lack the necessary skills, resources, and tools to implement portfolio management effectively. Traditional innovation portfolio management practices offer limited applicability to missions. Moreover, mission portfolio management requires flexible and adaptive processes to ensure its effectiveness, which often conflict with the rigid structures and practices typical of public administration. This paper introduces actionable guidance for the development, management, and monitoring of mission portfolios. It also provides targeted recommendations for adopting mission portfolio management in a purposive and dynamic manner.

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The project aims to enhance the evidence base on national mission initiatives and provide practical inputs for their effective implementation. It focuses on three dimensions of mission implementation: mission governance, mission portfolio management, and mission monitoring and evaluation.

A specific STI Paper is dedicated to each of these three mission dimensions. This particular paper deals with mission portfolio management.

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Executive summary

Mission Portfolio Management (MPM) puts portfolio management at the core of missions, adopting a purposeful, system-wide approach to initiating, aligning, monitoring, and steering diverse activities to achieve specific mission objectives. Unlike traditional portfolio management, MPM is embedded within broader ecosystems, encompassing diverse organisations, activities, innovation stages, and societal objectives. As a crucial element of mission implementation, it enables continuous coordination, prioritisation, monitoring, and delivery. However, it remains a novel approach, with many mission practitioners still in the early stages of the learning process.

Transitioning from traditional innovation portfolios to mission portfolios

Traditional innovation portfolio management provides a foundation for MPM but requires significant adaptation. While innovation portfolio management focuses on optimising resource use, risk management, and performance monitoring within single organisations, MPM expands and strengthens this approach by broadening the portfolio's scope to include upstream research, downstream market deployment, and social innovations. It shifts the emphasis from financial outcomes or isolated innovation results to societal impact, ensuring that contributions from a diverse range of stakeholders are integrated and aligned with mission objectives. Traditional practices often remain static, relying on retrospective tagging and mapping of activities, whereas MPM demands a proactive, dynamic approach that can adapt to evolving mission needs and maintain real-time alignment with the mission's theory of change.

Leveraging the benefits from mission portfolio management

In principle, MPM provides several systemic advantages critical to mission success. By raising management above individual projects, MPM enables a holistic view of systemic effects and pathways. This practice provides mission partners with continuous intelligence, learning and adapting from the results of projects and activities. Based on this deeper and broader dynamic sense-making, they can make strategic decisions to prioritise and scale up activities that show promise and phasing out those that do not. Within the mission portfolio, MPM supports the coordinated exploration of diverse options for realising the mission (the “problem-solution space”) in a context of radical uncertainty. It also identifies opportunities to leverage complementarities among activities, ensure their coherence and optimise the use of resources.

Overcoming the challenges of mission portfolio management

Despite its promise, MPM faces several challenges:

- Rigid public sector frameworks often impede the flexible and adaptive processes required for MPM.
- Designing dynamic portfolios while maintaining transparency and competitiveness poses a “proactivity vs. fairness” dilemma.
- Many mission-relevant activities lie outside the direct influence of mission managers.
- Effective MPM demands technical, strategic, and cross-sectoral competencies, often scarce in traditional public administrations.
- Budgetary and staffing limitations can hinder proactive and continuous portfolio management.

Developing, managing and monitoring mission portfolios

Based on insights from a variety of mission portfolio practitioners and experts, this paper analyses three interconnected components:

- Developing the mission portfolio – initiating and/or bringing together activities in a portfolio and strategically selecting and designing activities to meet the mission objectives in line with the mission's theory of change and/or strategic agenda.
- Managing the mission portfolio – proactive and continuous steering of the portfolio to ensure continued prioritisation and engagement with objectives as the mission develops.
- Monitoring the mission portfolio – continuous mapping, monitoring and sense-making of the activities making up the portfolio to ensure complementarity and reach and that the overall portfolio proceeds towards the mission objectives.

Enhancing mission portfolio management practices

To operationalise MPM effectively, this paper proposes several recommendations.

Developing the mission portfolio

1. Ensure that the mission theory of change drives the continuous development of the portfolios and broadens research and innovation activities to address systemic challenges.
2. Develop explicit criteria for the systematic development of portfolios based on the mission theory of change.
3. Distinguish the different layers of activities that are directly within the control of the mission from those contributing to the success of the mission, but not under the mission's 'sphere of strategic influence'.

Managing the mission portfolio

4. Perform portfolio reviews and stage-gate portfolio mechanisms to create strategic flexibility and avoid portfolio lock-in.
5. Foster a culture of trust and experimentation and ensure risk acceptance in portfolio processes.
6. Establish clear portfolio management responsibilities within the mission management team.
7. Promote interlinkages and exchanges between the different projects and activities within the portfolio.

Monitoring the mission portfolio

8. Monitor the mission's success based on the portfolio's contribution to realising the mission goal, not the aggregation of individual projects' success.
9. Develop specific visualisation tools to monitor mission portfolios.
10. Integrate monitoring with learning processes to ensure continuous improvement.

Conclusion and next steps

MPM is a challenging practice that integrates diverse actors, activities, and objectives across boundaries into a coherent and adaptive system to realise the mission goals. It aligns systemic efforts with societal goals, fostering innovation and collaboration. However, realising its potential requires overcoming institutional inertia, resource constraints, and capability gaps. Future efforts should focus on refining MPM tools and methodologies, fostering institutional capabilities, and building shared languages and frameworks across sectors. Collaborative initiatives like the Mission Action Lab can support practitioners in adopting MPM practices effectively, ensuring missions deliver on their ambitious societal objectives.

1 Introduction

This section explores Mission Portfolio Management (MPM) as a strategic approach to coordinating diverse projects and activities to achieve mission objectives. It examines key benefits and challenges, draws insights from innovation portfolio practices, and proposes a framework for developing, managing, and monitoring mission portfolios, offering actionable recommendations for policymakers and practitioners.

Portfolio management is an emerging topic both in innovation management in the public sector and more recently in connection to mission-oriented innovation policies (MOIPs). The practice of Mission portfolio management (MPM) consists in ensuring that a consistent and tailor-made set of diverse projects and activities contribute together to the realisation of the mission objectives. While they do not solve all mission coordination challenges, portfolio management practices are an integral part of the toolbox in various missions, under different forms. Successful portfolio management is seen as particularly important for tackling missions that require system-wide action, as well as for anticipating and adapting to change in real-time. It is used to maximise and keep track of the contribution of each initiative and its projects and activities to the overall mission, support exchanges and leverage synergies between initiatives, and build clarity around the intent of innovation and experimentation among an ecosystem of actors connected to the mission.

However, existing tools and methods for innovation portfolio management stemming from public and private research organisations are only of limited use in terms of their direct application in a mission context. Such tools tend to concentrate on maximising financial returns, managing individual projects of comparable nature and are mostly based in single organisations. They also focus on shorter timelines that differ from the context of missions. MPM should cover the breadth of actions under multiple partners (and the levers of change available to them) in the mission ecosystem, while assuring the continued validity of the overall theory of change and final achievement of the mission target. The current practice of portfolio management in the context of mission itself falls short of the guiding principles. It often consists of a rather static tagging of supported and related activities without the expected purposive steering, promotion of exchanges and monitoring against the mission theory of change. Many missions currently lack the necessary skills, resources, and tools to implement MPM effectively. Furthermore, the organisational and institutional framework conditions in which mission operate are often not conducive to MPM, which requires flexible and adaptive processes to ensure its effectiveness.

This paper seeks to contribute to the rapidly developing practice of MPM by framing it in the context of the mission concept at large as well as that of established portfolio management practices in the public sector. The purpose of this paper is to enhance the evidence-base for the effective leveraging of portfolio management in order to provide actionable advice on MPM to mission managers. To this end, the paper first frames the concept of mission portfolio management, analysing which learnings from innovation portfolios could be applicable to MPM. Then the paper addresses both the expected benefits and challenges in adopting a mission portfolio management approach in current public administration settings. Based on these insights, a framework for establishing MPM practices is presented involving key insights for the development, management and monitoring of mission portfolio. The paper concludes with concrete recommendations for MPM practitioners and guidance on developing potential tools for MPM.

2 Framing mission portfolio management

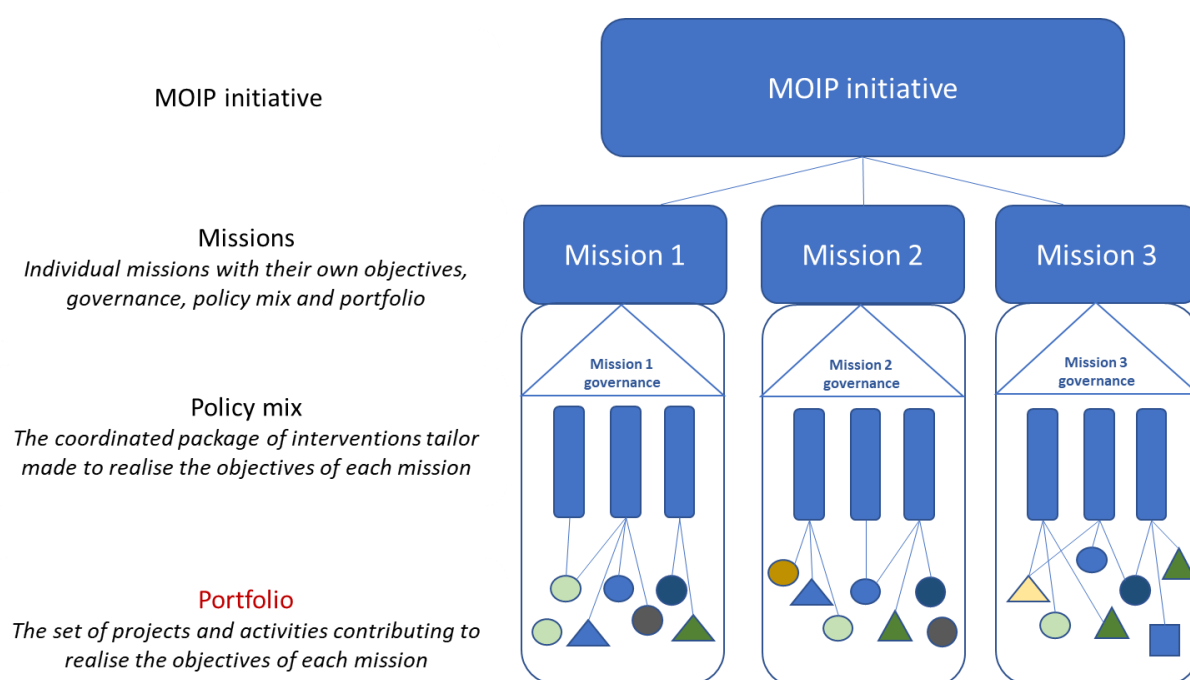
This section first positions portfolios within missions, then describes the progressive development of this concept, from 'traditional' innovation management portfolios to MPM, and from static to proactive MPM.

Positioning portfolios at the core of missions

Mission portfolio management (MPM) is the application of portfolio management practices to systematically initiate, align, monitor and steer a set of activities leveraged to achieve a specific mission. At the core of this practice lies the mission portfolio itself, that is the collected set of projects and activities that are purposefully selected, supported and combined so as to contribute together to realise the mission objectives. Similarly, mission portfolios were recently defined in a European Project as groupings of complementary actions (going beyond research and innovation, R&I) that are managed in a system-wide way to achieve the mission's goals (European Commission, 2024^[1]).

Figure 2.1 presents a simplified MOIP initiative with three missions. Each mission sets the goals and targets and mobilise a coordinated suite of policy and regulatory interventions (the policy mix) that support the various projects and activities (the portfolio) which together contribute to the mission realisation.

Figure 2.1. Main components of a MOIP initiative with three missions



Innovation portfolio management as a precursor to mission portfolio management

On initial examination, many aspects of innovation portfolio management, both in theory and practice, are relevant starting points for developing mission portfolio management. The general functions of innovation portfolio management are knowledge creation and sense-making, mapping portfolio activities and identifying possible synergies, ensuring agility across different activities, facilitating risk assessment across the portfolio, measuring and evaluating the innovation process, and ensuring effective activity co-ordination across the portfolio (OECD, 2022^[2]). Based on the functions of innovation portfolios, portfolio management tools should illustrate the distribution of resources and activities, promote a better understanding of portfolio activities, and help make sense of how they relate to often complex innovation systems (Table 2.1).

Table 2.1. Innovation portfolio management functions

Function	Role	Example
Sensemaking	Better overview and understanding of the current portfolio of initiatives and their synergies; optimisation of resources; strategic dialogue	Climate KIC uses portfolio approaches to create a shared understanding between partners to mitigate and adapt to climate change (e.g., in Transition Cities project) UNDP Portfolio Sensemaking and Acceleration Protocol ¹
Mapping and visualisation	Mapping innovation initiatives based on key criteria (e.g., connection to theory of change)	The provincial council of Gipuzkoa in Spain under the Building the Future programme has initiated the mapping of innovation portfolios to create shared understanding around innovation efforts among diverse stakeholders In the space of SDGs CGIAR ² has developed a dashboard for initiatives based on innovation readiness levels in specific intervention areas
Diversification	Dealing with uncertainty, spreading risks and avoiding lock-in	The US government's 10x agency, which funds innovation exploration activities and uses stage-gating funding to learn and assess exploitative potential over time The city of Helsingborg has adopted an innovation portfolio approach to track its innovations under its 17 challenges, meanwhile encouraging possibility-driven innovation in areas such as blockchain or Artificial Intelligence
Monitoring and evaluation	Portfolios contribution to the mission at hand; stage-gating investments and culling projects ³	The city of Lund uses an Innovation Ecosystem Portfolio Tracking (LIEPT) model to monitor innovation across the ecosystem The City of Chicago Office of Innovation uses a database to track ideas, progress and failures and provides a systemic overview of innovation progress
Portfolio co-ordination and stewardship	Aligning strategic actions across the portfolio	OECD's Portfolio Exploration Tool and the Innovation Portfolio Service help analyse innovation portfolios based on action-oriented criteria

Source: Based on (OECD, 2022^[2]).

However, existing innovation portfolio management tools only partially perform these functions. As portfolio management methods often originate from the financial sector with a view to steering investment decision-making to spread risk and/or maximise returns, they are commonly associated with evaluating investment options. Often, the end goal is to maximise the monetary value of the portfolio as a whole, achieve a

balance of projects across pre-identified criteria (e.g., exploration, exploitation) and ensure that the innovation portfolio reflects the organisation's strategy.

From innovation portfolios to mission portfolios

Therefore, innovation portfolio management practices and tools are not inherently suitable for missions. First, mission portfolios are broader in scope than innovation portfolios. The strategic orientation of a mission and its respective activities towards the mission statement is prioritised in mission portfolios; innovation, thus, serves that purpose rather than drives it. This means that mission portfolios must include both upstream and downstream activities that enable new solutions to have enough scale to achieve the mission, rather than only dealing with innovation funding. As such, a mission portfolio can be composed of projects and activities that span i) different components of the innovation system (from knowledge and innovation to skills, infrastructure, etc.), ii) different stages of development (from basic research to market deployment) and iii) different types of solutions (OECD, 2024^[3]). When designing and managing their portfolio, mission practitioners must consider a wider variety of criteria when mapping and making sense of the portfolio.

Second, the end goal of traditional innovation portfolio management, often driven by financial and risk-containment rationales, is too narrow. Achieving the mission, which is to address a complex and ambitious societal challenge, should be the primary criteria and focus for missions, with cost-effectiveness, feasibility, expected impact, etc., being assisting variables that help categorise activities. Since missions have wide and cross-cutting scopes, analysing the contributions of various actors to the mission portfolio is essential.

Third, mission portfolios involve several layers of activities contributing in different ways and to various degrees to the common objectives. Mission managers must engage not only with the activities and projects directly under their control (i.e., that they steer), but also with the activities and projects in the wider ecosystem that contribute more indirectly (or at least less formally) to the mission success.

From static to proactive mission portfolio management

Innovation portfolio tools and practices often fail to enable continuous and proactive management so new tools and methods are needed. Currently, the same holds for many missions. Often mission portfolio management practices are static, involving tagging and mapping activities relevant to the mission in specific sense-making exercises. More advanced software is in use (for instance, at funding agencies), but these are often case management systems or data visualisation tools in which the portfolio management is often limited to static efforts and lacks proactive and continuous deployment. Activities are ex-post “tagged to” particular missions based on their relevance, which helps to make sense of the operational context and to raise mission awareness and a sense of belonging. However, it does not allow for dynamic stewardship or coordination of the mission portfolio (see Table 2.2 regarding portfolio management functions).

Figure 2.2. Static and proactive mission portfolio management



Source: OECD authors.

In the context of missions, MPM should be a proactive practice (see Figure 2.2) that drives key decisions regarding activity selection, funding, connection to other portfolio activities, reorientation, termination, and more. It involves a dynamic sense-making and decision-making process for purposively building the portfolio and ensuring coherent resource distribution between activities. Mission portfolio management also entails providing stakeholders with continuous intelligence to facilitate learning and adaptation based on the results of projects and activities, and strategic decision-making that prioritises promising activities and phases out those that do not. A portfolio approach to missions also anticipates both the diversity and multiplicity of actions needed to achieve a mission.

3

The expected benefits and challenges of mission portfolio management

This section explores the expected benefits and challenges of mission portfolio management (MPM). It highlights how MPM enables greater coordination, strategic alignment, and systemic oversight of mission activities, fostering more effective decision-making and resource allocation. At the same time, the chapter addresses key challenges, such as navigating cross-sectoral complexities, ensuring adaptive governance, and balancing flexibility with accountability.

The expected benefits of mission portfolio management

The strengths of the MPM approach resides in the ability of mission leaders to design, steer and monitor the mission activities at higher and systemic levels rather than taken in isolation. This holistic and proactive management of projects and activities has several expected benefits.

Raising the level of management above individual activities to capture the systemic effects.

A project-centric view of innovation especially connected to missions can be problematic, and public sector organisations have been found to struggle with “projectification” – the linear division of policy problems into smaller, manageable, time-bound actions without a general overview of their collective impact (Hodgson et al., 2019^[4]). In the context of missions, more specifically, this holistic level of management is the one at which it is possible to practically connect these activities and the mission strategic agenda, i.e. the different pathways that are envisioned (and revised) by the mission partners to achieve the collectively defined objectives (see Box 3.1).

Box 3.1. Mission portfolio strategy in Japan’s Moonshot R&D Programme

Japan’s Moonshot R&D Programme has set seven very ambitious Moonshot goals such as the “realisation of a society in which human beings can be free from limitations of body, brain, space, and time by 2050”.

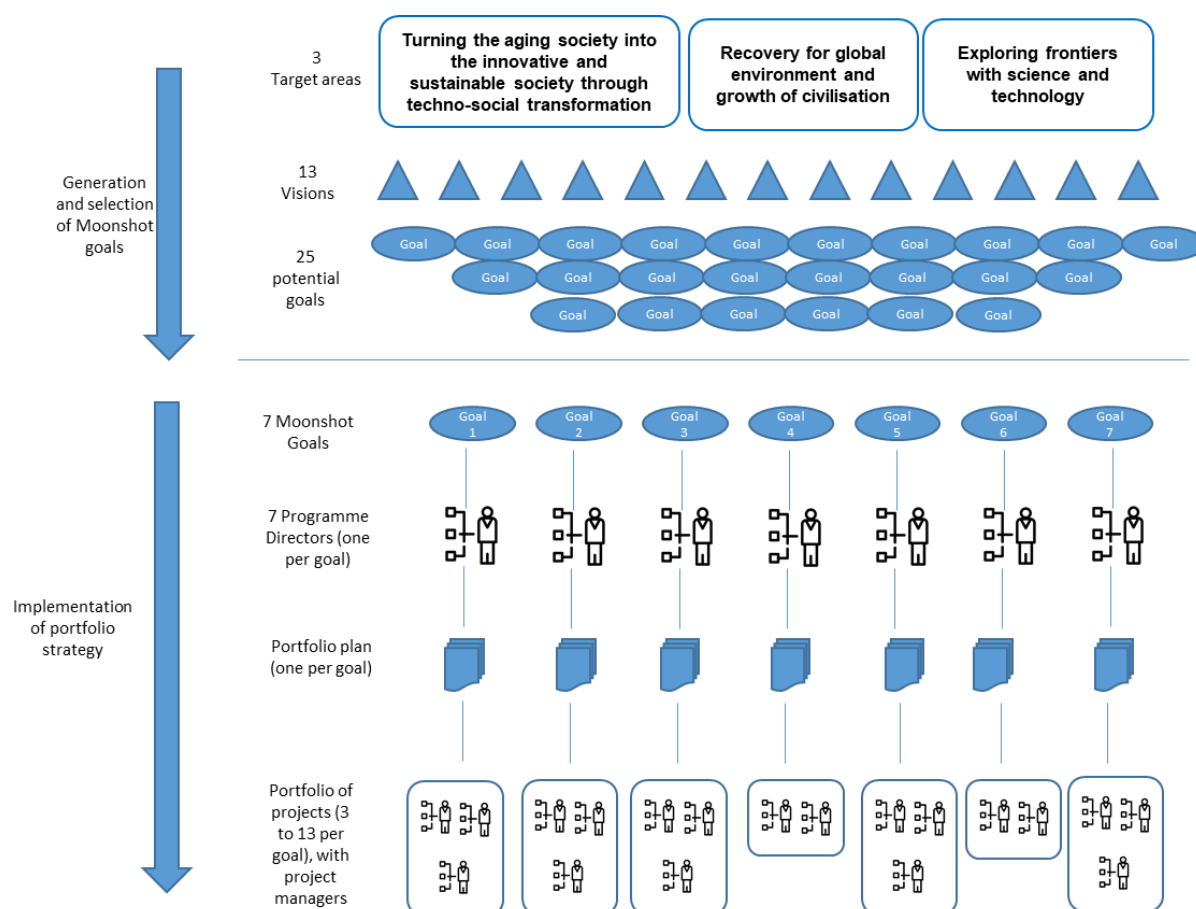
In each project, it funds frontier research focused on the achievement of these goals. At the level of each goal, it also implements a portfolio approach to manage the high risk of failure of projects. The main idea is to create an R&D portfolio system as a package of projects for each Moonshot Goal and evaluate the possibility of achieving a goal at the level of this package, not at the level of each individual project. Between 3 and 13 projects were selected for each of the seven goals (Figure 3.1). This strategy is reflected in the design of the programme. A Programme Director is appointed for each Moonshot goal and tasked with the development and implementation of a Portfolio

Plan. Programme Directors are asked to choose projects that explore different paths and methods to meet their respective Moonshot goals.

More generally, the Moonshot R&D Programme implemented an exploratory approach at all stages, from the selection of goals through brainstorming and consultations (starting from 3 target areas to open up to 13 visions and 25 potential goals, in order to finally select 7 goals) to the implementation of a portfolio strategy by Programme Directors.

Source: Larrue (2021^[5])

Figure 3.1. Exploration of potential problems and solutions in Japan's Moonshot R&D Programme



Enabling the coordinated exploration of a set of uncertain potential solutions.

Traditional portfolio management reduces the level of risks of a pool of projects when taken together. Supporting several projects following different paths maximises the chances that at least one project succeeds. MPM go beyond this by coordinating different actors in the innovation system and beyond to explore different options in parallel until sufficient information is collected regarding their respective merits. While these options are all relevant to the particular challenge the mission addresses, they can involve more than exploring solutions. They can also experiment with different pathways and markets, which often have to do with different views on the relevant problem itself.

This ‘coordinated exploration’ strategy is critical when the level of uncertainty and/or the level of contestation of the ‘problem-solution’ space of a mission (Wanzenböck et al., 2020^[6]) is high. Such an approach was, for instance, used in the Norwegian Pilot-E initiative (see Box 3.2).

Box 3.2. Coordinated exploration of potential solutions in Pilot-E

Within the same call procedure, although in the end the partners intend to select the projects that best respond to the standard criteria they have set, they tend to explore the technological landscape and fund different options. The first call funded, for instance, 15 consortia, including battery-powered ferries and hybrid hydrogen/battery ferries. They also deliberately cover different market segments, from different types of passenger vessels (urban water vessels and others) to cargos and ferries of different sizes. The second call was purposively broad to cover various emission-free land-based transportation segments (light and heavy-duty freight transport; Zero-emission construction machinery; Zero-emission public transportation).

From one call to the other, the ambition level has increased. While the first call for zero-emission maritime transportation did not specify other requirements than emission-free, the third call explicitly sought projects that commit to vessels that can achieve higher performance in terms of for longer distances, at higher speed and/or for longer duration (‘Emission-free maritime transportation 2.0’).

Finally, Pilot-E also follows a value-chain strategy. As part of the first call, Pilot-E funded not only projects for zero-emission boats but also the battery power charging system. The third call also funded Zero-emission maritime transportation and infrastructure for zero-emission ships, and the fourth call funded hydrogen supply chains.

Source: OECD MOIP Online Toolkit, <https://stip.oecd.org/moip>.

Identifying and leveraging complementarities between the various projects and activities within the mission portfolio

Projects within a portfolio do not perform in isolation, they form part of a broader organisational or systemic context. Complementarities in missions can take the form of either, (1) technical/substantive synergies between the different activities/projects in the portfolio or (2) more efficient management of resources. This can result in better upstream and downstream management of mission RDI and the pooling of resources to achieve the mission. It is expected that public authorities and the coordinators to whom they delegate the task of orchestrating the mission ecosystem are expected to play a key role in incentivising and promoting exchanges within the portfolio to maximise and leverage ‘technical’ complementarities between interdependent activities pertaining to the same set of potential solutions to the mission.

Two different types of technical/substantive complementarities can be identified:

- Complementarities between activities within a given set of ‘innovation solutions’: These are the complementarities between activities pertaining to a specific solution (i.e., a specific socio-technical system or pathway). For instance, developing the production and commercialisation of carbon-neutral hydrogen requires the development of new technologies of electrolyzers, the deployment of new infrastructures for the distribution/transport of hydrogen, possibly a capacity to produce renewable energy for producing energy, the development of new engineering skills to produce and maintain the electrolyzers, etc. The same applies to many net-zero systems. For instance, a net-zero farm will require new methods (e.g., for carbon-neutral fertilisation, for increasing farm efficiency), new technologies (e.g., renewables, carbon capture systems, sensors), as well as activities along the value chain, including buyers and customers. These different components of the systems are interdependent, and the mission’s success will depend on the consistent progress of all of them.

- Complementarities between solutions at different stages of development: A mission often supports different generations of solutions, from the more exploratory to those more advanced and closer to market. Significant lessons learned are drawn from the early generations and fed into the development of more ambitious but still underdeveloped solutions. There are, therefore, interdependencies between activities at different stages of development (from exploratory research to scale-up and market deployment) for achieving the same mission in the longer term, which covers the different stages of the innovation chain.

Harnessing these complementarities through proactive and purposive management is beneficial to achieving the mission. Mission portfolio managers and their partners must be able to identify the different types of complementarities and, therefore, needed linkages—between activities within a given mission portfolio. A systemic approach to MPM is essential to ensure the various intervention points are working in a coherent manner across the missions they are trying to influence.

Portfolio management can also be instrumental in linking and aligning mission activities with other relevant initiatives of similarly minded initiatives, as illustrated in the case of the Viable Cities mission (Box 3.3).

Box 3.3. Managing the climate transition through a portfolio approach – Viable Cities, Sweden

Viable Cities is a strategic innovation programme (SIP) hosted by the Royal Institute of Technology (KTH) with support from the Swedish research and innovation funding agencies Vinnova, the Swedish Energy Agency and Formas. Their funding amounts to EUR 100 million over 12 years and is, to a large extent, dedicated to grants. The programme is focused on making cities climate-friendly and sustainable. The strategic innovation programmes are one of the main mission-oriented innovation policies that have been implemented in Sweden for the past 10 years.

Viable Cities adopted a portfolio approach to achieve a systems view of the climate transition. The portfolio approach helps to counter siloed and fragmented efforts that do not contribute sufficiently to climate transition. The programme identified a “projectification” mindset – dividing problems into discrete linear problems and setting up individual projects to tackle them – as a key obstacle to successful innovation work. For this reason, portfolios are not conceptualised as a list of projects; instead, they include everything seen as relevant to the mission, including challenges, needs, barriers or future options. Further aspects may include investment plans, learning and people.

The portfolio approach encourages member cities and external partners to co-develop a strategy. It is hoped that portfolios with an “open interface towards other portfolios” will create a common language and framework and thus facilitate discussions and exchange of ideas. This way, the portfolio of Viable Cities may become compatible with those of similar organisations. However, ensuring compatibility among portfolios requires understanding the different international, national and municipal institutional levels at which they operate. Viable Cities focuses on city-level innovations where concrete implementation measures are taken.

Source: OECD (2022^[2]).

Mission portfolio management challenges

While MPM is a crucial dimension of missions, it is not implemented in all missions. It is new and presents many challenges for public authorities. Most missions adopt only part of this approach and are still experimenting with new methods and tools. In many respects, MPM is still very much a learning process, with little information available on successful practices.

Setting the right framework conditions

MPM necessitates the creation of a novel and agile work framework, which is often not possible within the current innovation funding and policy programme structures. This framework notably includes different types of criteria for selecting activities, processes to steer and monitor activities, internal (administrative) incentives, and rules to align staff with portfolio management logic.

The setting of criteria to select activities and design the portfolio is particularly challenging. Traditional management often uses criteria related to research excellence and relevance. MPM relies upon these criteria as well to assess the proposed projects but adds an essential new family of criteria at the portfolio level: the coherence of the portfolio relative to the mission objectives, i.e. how each project contributes to the mission objectives as part of a consistent set of projects and activities. However, when mission portfolios become very large, then there also needs to be a cutoff point in terms of which activities and projects are considered in active MPM (based on critical size, expected impact, etc.). The operationalisation of this type of 'portfolio fit' criteria is not yet well formalised.

Combining proactive management with fair competition

How to proactively design the portfolio while still preserving a sufficient level of competition between initiatives and ensuring a sound and equitable treatment of project proposals? Relying upon 'distant' calls for proposals ensures a level playing field between different candidates and keeps the process accountable. It is less the case when using more hands-on approaches, which existing government or organisation-level administrative rules might even forbid.

In most cases, public administrations can only allocate funding using competitive mechanisms (e.g., calls for proposal, tenders and public procurement), which do not easily allow for selecting specific projects/activities based on their complementarity with other projects in a portfolio.

Another challenge related to the relationship between the mission approach and market competition concerns the ways to promote exchanges between different projects while preserving the confidentiality of certain knowledge and information. An essential dimension of portfolio management is promoting exchanges between complementary activities within a portfolio to leverage potential synergies. However, project participants – especially in private firms and Research and Technology Organisations (RTOs) – can sometimes be reluctant to exchange with other organisations for fear of disclosing proprietary information. Monitoring the portfolio in the collective setting of the mission can also face similar confidentiality issues.

Managing activities that are 'out of control'

As previously mentioned, the portfolio often includes activities that lie outside the mission managers' direct control but are relevant to the mission's progress and accomplishment. Therefore, the relation to these different activities will be different, from direct coordination in the case of activities within their control to only information exchange for more 'remote' activities. This is an essential parameter to consider. Although missions are meant to be systemic and involve a broad array of public and private actors, mission managers are often located in one part of the system, i.e., a specific administration or platform. Their power to manage activities, therefore, depends on the nature of these activities. They often face difficulties managing activities that pertain to a policy field or a thematic area outside their organisation's mandate.

A recent OECD study in the area of missions to combat climate change shed light upon the consequences of their partial control over their portfolio (OECD, 2024^[3]). As most missions are initiated and managed by public authorities in the research and innovation area, portfolio management can only be proactively exerted on the activities they can mobilise and support with their set of interventions. In this view, a mission portfolio is often better understood as composed of nested concentric circles. At the core of the mission

portfolio lie the activities directly governed by the mission (primarily those in the research and innovation area). The portfolio's wider circles include activities less influenced by the mission's common strategic agenda (mostly activities in sectoral policy areas). At the very periphery of the mission portfolio reside the activities that are relevant to the mission's challenge but are not connected to it in practice, i.e. they have been initiated, programmed and implemented in isolation from the mission decision-making processes (Larrue, Tönurist and Jonason, 2024^[7]).

Many so-called net zero missions remain limited to research and innovation activities, particularly relevant technological options, while reducing emissions calls for changes far beyond this sphere, notably behavioural and social change.

Nurturing the necessary capabilities and tools to manage strategically the portfolio of interventions

Many public administrations can be in limited supply of the needed skills and knowledge, as they traditionally prioritise administrative and process-oriented skills over substantive and strategic skills. Depending on the mission, MPM can, for instance, call for knowledge of specific technological and industrial areas to plan and assess the portfolio's progress towards the objectives, identify arising needs, and terminate underperforming or counterproductive activities. It also seeks to analyse and understand how each project or activity, as part of the broader portfolio, contributes to the mission.

This demands some 'T-shaped' competencies that are not only deep but also broad since, as previously mentioned, the portfolio might include activities of different natures in different sectors and cover different stages of the innovation chain.

As most missions have complex project and activity portfolios, mission managers must have access to or develop their own MPM tools. For this relying on a concise theory of change is a must to identify specific, measurable and relevant indicators to analyse portfolio activities continuously. Measures to quantify the level of resources and potential impact may be required to make portfolio-level decisions. Portfolio management also differs from traditional project management because it requires the analysis of risks and impacts on the portfolio level, and usually, this is not done in the public sector, potentially necessitating the reskilling of mission managers and teams.

Managing the portfolio at a reasonable cost.

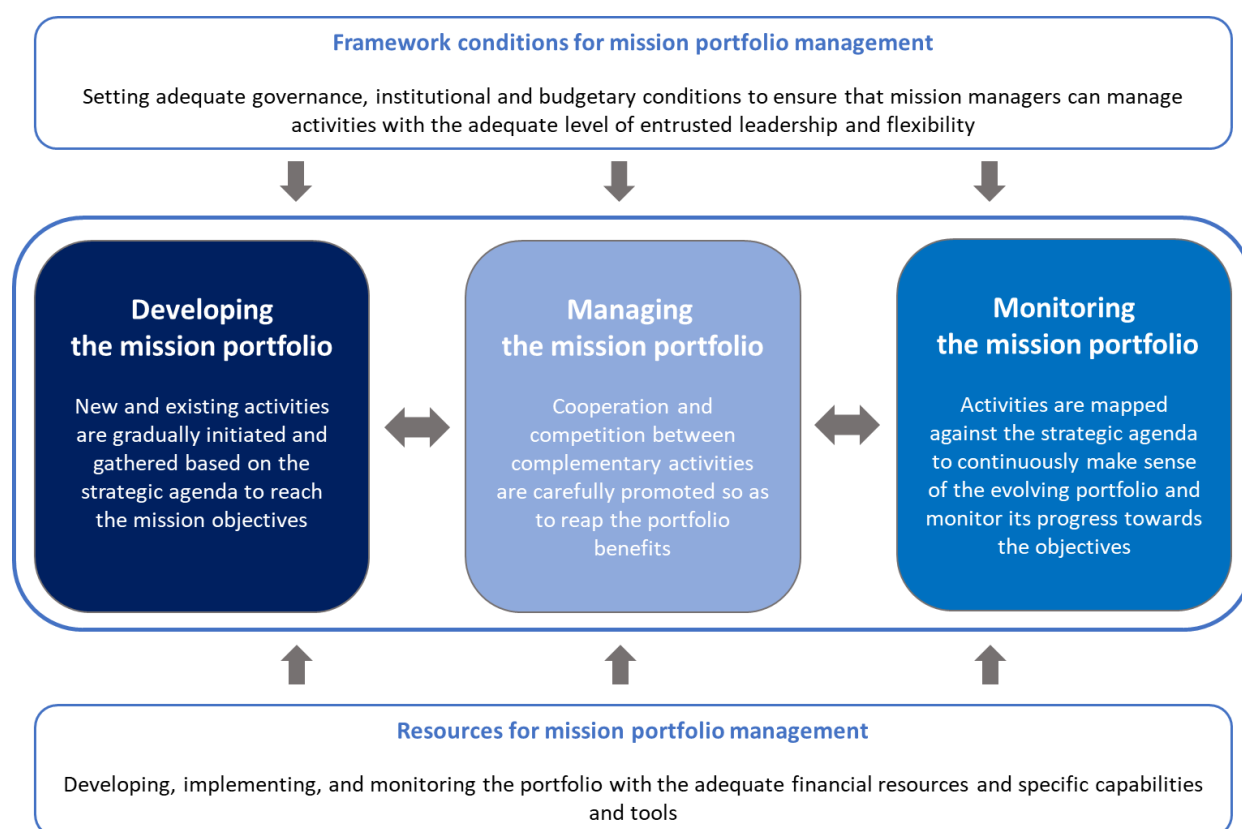
The challenges set out above make clear that MPM requires increasingly costly human resources. These practices require additional budgetary resources in public administrations (notably agencies) and various institutions serving as coordinators/orchestrators of the mission. These resources are difficult to secure in administrations subject to budget austerity and strict staff limitations after years of New Public Management.

4 Mission portfolio management practices

This section introduces a three-pronged framework for mission portfolio practice, drawing on insights from a diverse group of practitioners and experts. These perspectives were gathered during the project's benchmarking phase through interviews and a knowledge exchange workshop:

- Developing the mission portfolio – initiating and/or bringing together activities in a portfolio and strategically selecting and designing activities to meet the mission objectives in line with the mission's theory of change and/or strategic agenda.
- Managing the mission portfolio – proactive and continuous steering of the portfolio to ensure continued prioritisation and engagement with objectives as the mission develops.
- Monitoring the mission portfolio – continuous mapping, monitoring, and sense-making of the activities making up the portfolio to ensure complementarity and reach and that the overall portfolio proceeds towards the mission objectives.

Figure 4.1. Components of mission portfolio management



Source: OECD authors.

The effective deployment of each component depends on appropriate framework conditions, namely adequate institutional and budgetary conditions that secure dedicated financial resources and specific mandates and capabilities. These are, in turn, in large part, results of the broader mission governance (see separate working paper). See an example from the UK innovation agency, Nesta (Box 4.1) on how these mission portfolio components are applied in practice.

Box 4.1. An illustration of the three components of portfolio management: the case of Nesta

Nesta is an independent foundation which bills itself as “UK’s innovation agency for social good.” The organisation, launched in 1998, has fully transitioned its operations to a mission-oriented approach in the last three years. It now focuses entirely on how innovation can be harnessed to address “wicked” social problems through the three “innovation missions” that it launched in 2021: a healthy life (addressing health inequalities), a fairer start (narrowing the education attainment gap), and a sustainable future (accelerating the decarbonisation of household activities). In this work, Nesta sees itself as playing three mutually reinforcing roles. In addition to being an innovation partner, Nesta seeks to act as a system shaper, working through the barriers that limit transformation and mission impact, and as a venture builder, carving out new paths with stakeholders and providing financial support to new and early-stage ventures.

Developing the mission portfolio

Nesta’s strategy is to build portfolios around each of their missions’ theory of change, which is informed by how Nesta best fits in the ecosystem and can provide the most impact through each of its three roles. The portfolio mix focuses both on the potential impact of individual projects, as well as the interrelation of projects and value-add of their effects taken together.

Managing the mission portfolio

Building towards a strategy that spans until 2030, Nesta is experimenting with different internal processes to regularly check in on their mission portfolios and adapt their theories of change as solutions, learnings, and challenges arise. Currently, each portfolio is reviewed by the executive leadership bi-monthly, looking especially at the “biggest bets” of each portfolio. The theory of change is revisited about every three years.

Monitoring the mission portfolio

Each mission has a portfolio of circa 15-20 projects and activities, which are continuously mapped and managed in the digital platform Asana. Each project or activity is tagged along multiple dimensions, such as area of focus or innovation stage. Metrics are continuously developed to capture the impact of projects and the broader portfolio.

Note: This case highlights an interview with Hasan Pesaran, Senior Strategy Manager at Nesta, conducted in November 2024.
Source: Nesta (2023^[8]).

The above three-pronged structure of mission portfolio practice was used to structure the exchange with various practitioners to distil practical insights on the operationalising of each component.

Developing the mission portfolio

Missions do not necessarily need an extensive and diversified set of projects (even though tackling complex challenges generally involves many approaches); they do, however, require a clear and coherent strategy on how each of their activities and projects help reach laid-out targets. Developing and building a

systemic and proactive mission portfolio should be informed by a coherent theory of change and leverage points that will encourage buy-in amongst stakeholders. The traditional criteria for selecting activities themselves need to be changed, particularly to factor in the consistency and complementarity among themselves and with the objectives and theory of change. Collective sensemaking helps identify root causes, barriers to change, and possible interventions using a developing shared vocabulary. It also builds trust and legitimacy as the mission portfolio moves from an untested theory of change to tangible on-the-ground projects and activities.

In this line, the following key questions were raised with mission managers and experts:

- At what level of mission governance should mission portfolios be developed? How to structure “nested” mission portfolios (activities and projects of organisations contributing to the mission, but not leading it)?
- Which frameworks (strategic agenda, theory of change, intervention logic, portfolio plans, etc.) should be considered when creating and organising a portfolio? What are the challenges and good practices in using these models?
- What criteria can be used to “populate” the mission portfolio and reflect the consistency/coherence of mission-oriented activities? To what extent can traditional criteria (e.g., excellence, relevance, closeness to market/scalability, feasibility, returns on investments) be used in the context of missions?

In responding to the questions above, mission practitioners found that the selection of projects should primarily be based on their relevance and potential for the transition pathway of the mission. As one interviewee puts it, “everything flows from the theory of change.” For practitioners, mission portfolio management structures should be built with a wide range of stakeholders, mapping and engaging with the whole mission ecosystem, both vertically and horizontally. Additionally, addressing the nested structures of portfolios from the get-go to account for contribution effects is important. As horizontal and vertical policy integration is needed, the different ecosystem partners will select “their portfolios” from an overarching one and complement them with their stakeholders’ actions (by contributing them to the broader mission portfolio). This dynamic should be clear to the central mission manager and the key stakeholders expected to contribute to the mission portfolio.

Due to limited resources and clear timeframes, practitioners often find themselves balancing ambition with practical considerations. The temporal aspect of missions has been pointed out as necessary for formulating criteria at every mission stage, aiding in identifying underdeveloped areas within the portfolio. Practitioners found that the concept of ‘scale’ is often not valuable for multi-organisation mission settings, and the willingness to engage with the mission should be described in other terms (e.g., willingness to transform). For example, if a mission depends on external funding, the sustainability of the latter should be taken into account, and the necessary partnerships should be built into the theory of change (ToC) underlying the establishment and design of the mission. There are tools available to support the creation of an effective ToC.⁴ When ToC and mission portfolios are closely aligned, MPM, in practice, also creates the conditions for partnerships to guide organisational transformations.

Box 4.2. Portfolio management in Austrian transition missions

The BMK’s approach to portfolio management focuses on impact pathways, outlining the desired outcomes of research and innovation investments in key areas like climate neutrality, circular economy, and energy transition. This approach helps to avoid duplication of efforts and ensures that different activities complement each other. The Austrian Research Promotion Agency’s (FFG) work program is central to operationalising these practices. It manages various funding programs and initiatives aligned with the BMK’s impact pathways. The work program includes calls for proposals designed to

complement other funding opportunities and are often co-financed with other ministries or regional governments. The evaluation and selection of projects are based on several criteria including relevance to the call, scientific and technical quality, impact, implementation plans with timelines, resources and partnerships, the degree of novelty and innovation beyond the state of the art, as well as its alignment with the BMK's strategic objectives. For instance, in the Circular Economy Mission the enhanced coordination within BMK and FFG establishes continuous and systematic links between research and innovation measures and implementation support actions. This addresses a clear demand from the BMK Minister, enabling better alignment of investment funding, market deployment, awareness raising, education, and regulations with the Circular Economy Strategy.

The BMK holds regular review meetings to assess the progress of each mission. These meetings bring together representatives from different ministries, funding agencies, and mission platforms. They discuss the implementation status of various measures, identify challenges, and explore opportunities for improvement. These meetings provide a forum for exchanging information and coordinating actions, ensuring that the portfolio remains aligned with the mission goals and complements each other.

Source: OECD MOIP Online Toolkit, <https://stip.oecd.org/moip>.

Managing the mission portfolio

As the mission matures, the focus shifts to managing and steering the portfolio. Proactive decision-making is critical, as the mission managers need to be reactive to the problems and results of individual projects and activities, as well as the hypothesis testing and changing context of the overall mission. For this purpose, mission managers must frame negative results from specific projects as successful learning outcomes for the broader mission and re-engage stakeholders in other efforts when possible. This iterative learning process and culture is key to dynamic mission portfolio management, where projects and activities must inevitably be scaled up or down based on their progress.

In this line, the following key questions were raised with mission managers and experts:

- How can proactive portfolio management be enabled, in which projects and activities are scaled down or up based on their results as well as evolving mission context, learnings, and strategies?
- What decisions at the portfolio level should the mission manager be able to make? Would the previously discussed criteria allow those decisions to be made?
- How can exchanges and complementarities between the portfolio's instruments and activities be supported to leverage complementarities and maximise impact? How to take into account the possible competitive and confidentiality issues between various participants in the portfolio?
- How can dynamic portfolio management be established that encourages flexibility, minimises lock-in from prior commitments, and remains open to new, experimental, and/or transformative options along the way? Are there any adjustments to the portfolio criteria discussed that you would make keeping this in mind?

First and foremost, the mission practitioners found that it is essential that there are people or particular institutions that are seen as having the role to "manage the mission," part of their role is also to build and curate collective action (e.g., see practice in CSIRO – Box 4.3 below). Suppose the de facto mission managers lack the legitimacy to do so. In that case, it is very difficult to develop the capacity, skills and tools for managing a mission and taking a proactive role. Consequently, a key enabler is "assigning" or "recruiting" this role, which also helps assign mission portfolio managers and support them with common frameworks, processes, vocabulary, and digital tools. In some cases, for example, platforms of ecosystem-

led missions (see mission governance working paper), ecosystem orchestrators, or network backbone organisations might be more useful frames of reference for leading missions than the centralised role of mission managers. This type of role should be facilitative in terms of leadership and help the (eco)system see itself, govern itself and make decisions about itself.

Box 4.3. CSIRO's nested governance approach to portfolio management

CSIRO (the Commonwealth Scientific and Industrial Research Organisation) is an Australian Government agency responsible for scientific research. CSIRO uses a three-tier system of nested governance to manage its mission portfolios. This includes:

- the Work Package Leads manage a discrete portfolio of activities known as “work packages.”
- the Mission Leads manage the missions as a collection of work packages.
- the Missions Program Office manages the whole portfolio of missions.

Practically, developing and managing each of the above is an almost constant process of negotiating the interests and agendas of mission stakeholders (internal and external) and engaging with the feedback loops between mission activity, MEL, design, strategy and foresight.

Individual missions contribute to biannual progress reports for their entire portfolio, tracking their advancements against Impact Pathway indicators that measure progress toward defined environmental, social, and economic outcomes.

Source: OECD MOIP Online Toolkit, <https://stip.oecd.org/moip> and CSIRO (2023^[9]).

Once the mission manager role is established, concrete roles and processes for dynamic MPM can be implemented. For example, a monthly learning log to capture the insights in the learning experiments, collective sensemaking every 3-6 months to extract insights on the overall/specific effects, and making decisions whether to start, stop, continue or scale in the current activities. More often than not, active portfolio management might focus on the “next step” to drive the mission, less on managing existing projects’ portfolios. Nevertheless, mission practitioners were adamant that mission managers should have the flexibility to make informed changes in the current activities due to learnings and insights from collective sensemaking (or similar sort). At the portfolio level, the manager should be able to make decisions about initiating new actions, linking actions together, further investments/scaling up actions, and stopping further investments in actions.

Adopting a broader portfolio management perspective that extends beyond up- and down-scaling is crucial. Some practitioners illustrate this by highlighting instances where other actors replicated portfolio activities, translated them into legislative action, mainstreamed them, or evolved them into entirely new initiatives. Additionally, different stakeholders may view mission portfolios differently. While at the STI level, missions may be understood as a shared learning and experimentation environment, which stakeholders use to pick and pitch ideas, for mission managers, the goal is to find the tipping points, including the most impactful pathways to change in the portfolio that can help the delivery of the mission. In practice, this may mean applying thresholds for starting, scaling up or stopping actions within the mission portfolio while also providing “free funds” to act as seed money for more experimental parts of the mission portfolio. Hence, accountability in traditional terms also differs from MPM. Not every portfolio item should have a high evaluation burden; instead, ToC should be designed to give flexibility and accountability towards the mission, and not every individual output should be at the activity level. This also means that for MPM to work in practice, mission managers should have a license from senior management to engage in a more agile and iterative process. This may require senior executives to appreciate the intent of iterative design

to carve out the space for mission teams to engage meaningfully and without fear in the iterative design process.

Monitoring the mission portfolio

When mapping and monitoring the portfolio, it is essential to differentiate between the projects and activities directly under the control of the mission team and those in the broader ecosystem that are relevant to the mission but over which the mission manager has little to no agency. Once again, the mission's theory of change, established at launch and adapted throughout the mission process, is important in informing and guiding active and responsive monitoring. Continuous visualising the project portfolio – with simple means or more elaborated dynamic monitoring and visualisation tools – can assist in information exchange amongst stakeholders and support proactive decision-making to realise the mission objectives.

The following key questions were raised with mission managers and experts:

- How to ensure that portfolio monitoring informs both collective learning (on the mission ecosystem level) and the strategic agenda?
- What are the metrics of success by which the portfolio's progress is measured?
- What visualisation and decision support tools are best used to map and monitor a portfolio?

Mission practitioners found that mission monitoring should be carried out on many levels and at different cadences (cycles of time) that are relevant to the actors at each level. Systemic, strategic, operational, and tactical levels view activities and outcomes at different time scales. Often, this requires creating a matrix of levers of change and transition phases that would allow a layer of different mappings in order to identify synergies within the portfolio. Also, the ability to zoom in on different levels of portfolio activities (e.g., on a particular strand of intervention or geographical level (EU-national-regional) is important as well. While the wish list for mission monitoring is often quite long, first and foremost, it is essential to link the theory of change to monitoring, evaluation and learning (MEL) frameworks much more explicitly. In practical terms, the process may start by using unstructured tagging to identify projects that are either "missions aligned" or "missions originated". This gives organisations leading the missions the bare minimum of information required to interrogate the portfolio (insofar as it sits within this organisation). Having a clear connection between the ToC and practical portfolio activities makes it clear that different capabilities and attitudes also need to be exhibited by mission managers and contributors. Often, opacity suits managers and mission contributors as it allows them to continue business as usual – this is something that mission monitoring should help avoid by actively helping participants to also feel less threatened and more reflective.

While key KPIs in mission portfolio monitoring may differ across missions and approaches differ depending on the organisation (see Box 4.4 for an example from UNDP), broadly speaking mission practitioners are looking at the following dimensions against the counterfactual of a linear, siloed programme: acceleration (of impact delivery), orchestration (of system directionality), spillovers, mission configurations (of work packages), portfolio configurations (of mission types). These, in practice, may include cumulated investments, actor engagement/composition, and "tag" various categories of system effects and different phases of development.

Box 4.4. Emerging criteria in monitoring mission portfolios in UNDP

Since 2024, the UNDP has introduced a new delivery instrument on portfolio policy with various guiding materials on creating a theory of change for the portfolio, classifying projects and their risks. The

connected methodology does not set universal indicators for monitoring portfolios but outlines how a portfolio's theory of change key outputs (shifts instrumental to unlocking change) are connected to one or multiple potentially valuable indicators to help monitor and evaluate progress in the specific area. For this, the Iceberg approach to layering multiple indicators is suggested.⁵ As such, for example, the UNDP Regional Bureau of Eastern Europe and Central Asia often uses indicators that examine new/emergence (what is the application of learning and making changes), knowledge/partnerships, capabilities/skills, and behaviours/mindsets connected to their portfolios. These dimensions help UNDP track the portfolio's success in terms of outcomes and process-level changes.

Source: UNDP (2024^[10]) and UNDP (2022^[11]).

When mapping and monitoring the portfolio, practitioners look to develop a “knowledge infrastructure for networked governance.” Having specific portfolio tools (usually digital tools) to collect the data from different perspectives to get a holistic picture of what is happening is often very useful. Here, an essential concept for MPM is “portfolio views”, which help different types of stakeholders look at data visualisations that are relevant to them and use that to track progress and understand what is going on and what might be missing. Other methodologies, e.g., network analysis, are also options for understanding the evolution of portfolios over time and measuring the quality of the composition in terms of relational assets and knowledge creation, in particular, if the goal is to foster new topics and sectors. (Penna, Romero Goyeneche and Matti, 2023^[12]). The portfolio mapping and monitoring may be publicly available (to some level of granularity) to enable mission actors to identify and connect to potential collaborators (see, for example, the European Commission dashboard on Oceans and Waters⁶). Although graphic tools are convenient for displaying data across different metrics on missions, effective mission monitoring also requires some subjective level of interpretation. Hence, practitioners find it helpful to engage in reflection and sense-making sessions or compile concise reports that blend data with a narrative highlighting alignment with the mission strategic agenda.

Box 4.5. Portfolio monitoring in EU missions

The Commission is planning to adopt a systematic approach to cataloguing and mapping mission portfolios. The approach employs a matrix structure that systematically organises the diverse actions required to achieve a mission's objectives. The matrix comprises three dimensions: strategic strands of intervention (the mission's objectives, thematic areas, and geographical scope), levers of change (governance, finance, individual/collective action, science/technology, and education/skills), and phases of system transition (experimentation, stabilisation, diffusion/disruption, and institutionalisation). The multi-dimensional framework is designed to be flexible and adaptable to each mission's specific context and objectives to identify synergies, gaps, and overlaps between different actions and interventions.

An analysis of the portfolio of the mission on 'restoring our oceans and waters' (841 projects) was performed in 2023 following two main steps: firstly, a search using the Commission's text mining tool CORTEX, which covers information from more than 60 directly managed programmes, on the bases of 200 keywords; secondly, an expert assessment of a sample of projects selected from the overall pull retrieved by CORTEX, specifically those with the highest relevance score. This analysis shows that the portfolio approach has successfully supported projects that align well with the Mission's objectives. This is demonstrated by covering a broad range of thematic areas and objectives. The portfolio analysis also identifies gaps and provides recommendations for future implementation, including addressing the underrepresentation of freshwater ecosystems, improving alignment between different funding programs, increasing the focus on knowledge transfer towards deployment and commercialisation, and broader involvement of public and private actors.

Source: European Commission (2024^[11]); European Commission (2023^[13]).

Continuous mission monitoring should feed into collective learnings and insights exercises that could be a form of sensemaking with key mission stakeholders to discuss among them what has happened across the portfolio and develop actionable insights for the whole portfolio's mission and effects. Systemic knowledge creation and tying that into the practice and actual work of the ecosystem partners or network is essential here. Hence, mission monitoring information must be made accessible and understandable. The core team of a mission manager or mission platform should be able to act as an "interpreter" and "assistance for sensemaking". This does not need to be complicated in practice. Mission practitioners often seek tools to combine mapping, sense-making, and tracing learning logs. Similar off-the-shelf digital tools include collaboration tools such as Miro, project management tools such as Asana, or data visualisation tools such as Power BI or Qlik. A combination of processes (reflection/sensemaking sessions) and some way to visualise the information embedded into regular processes (e.g., board or management meetings, reporting processes) is core to effective mission monitoring.

An action tool for mission portfolio managers

The table in Annex B proposes a simple and pragmatic synthesis of some of the main insights on MPM presented throughout this paper. It is a first attempt that should be complemented and tested with mission and portfolio managers.

5 Recommendations and next steps

This chapter presents key conclusions and recommendations for mission portfolio management, emphasising the importance of cross-sectoral and cross-organisational learning. A critical challenge is ensuring that these efforts foster shared languages and methods for effective decision-making, ultimately enhancing the successful realisation of missions.

Fundamentally, when performed well, mission portfolio management is a continuous activity that spans different institutions across disciplinary, sectoral and administrative boundaries. This represents both a major challenge and a key promise of the practice. A critical question going forward is, thus, how to ensure that cross-sectoral and cross-organisational learning and capability building lead to shared languages and methods for effective decision-making that support the realisation of the mission. To help do that, the paper puts forward the following recommendations using the three MPM components.

Recommendations for developing the mission portfolio

- 1. Ensure that the mission theory of change drives the continuous development of the portfolios and broadens beyond research and innovation activities to address systemic challenges.**

In essence, the mission portfolio should not be driven by what the main partner organisations have at their disposal but by what the mission theory of change requires. Given the origins of the mission-oriented approach, it is frequent that governments rely on traditional science, technology, and innovation (STI) instruments, such as research grants, to implement missions. The design of these instruments (mandated objectives, scope of eligible expenditures, etc.) constrains the portfolio to certain activities and options, most often related to technological innovation. However, to capture the synergistic and transformative outcome of missions, governments should also harness other policy instruments - ranging from regulation and taxation to procurement, scale-up mechanisms and incentives, communication, or education – in order to populate the portfolio with activities identified in the mission theory of change (or strategic agenda, pathways, roadmap, etc.).

- 2. Develop explicit criteria for the systematic development of portfolios based on the mission theory of change**

The criteria for selecting activities that will constitute the portfolio should be mission-specific and include variables on expected contributions to achieving the mission objectives, level of experimentation, market readiness, scale, etc. These criteria should be specific enough to initiate or select activities that can fill gaps in the mission theory of change, enable monitoring of whether the overall mission portfolio is likely to deliver on the mission objectives and cut activities that are not contributing to the overall mission continuously. These types of ‘portfolio fit’ criteria (i.e. assessment criteria that take into account how the different potential activities would contribute to the overall mission goal alongside the other activities in the portfolio) should be explicit in the ‘mission operation guidelines’ (or related intervention guidelines), along with and at the same level as traditional excellence, relevance and team experience criteria.

3. **In developing and managing the portfolio, distinguish the different layers of activities, from those directly within the mission's control to those contributing to the mission's success but not under the mission's sphere of strategic influence.'**

From the outset, the mission portfolio should be designed based on a clear understanding of which activities are within the control of the mission management team (operational environment) and which are important, critical activities contributing to the success of the mission but not under the mission 'sphere of strategic influence'. Recognising this distinction is particularly important given that many missions operate within a limited scope of activities compared to what is required to realise the mission entirely. This understanding supports two key objectives: i) ensure a comprehensive strategic outlook as it helps the mission managers and partners to move beyond a limited focus on their immediate responsibilities and constraints (avoiding the "half-empty glass" perspective); ii) actively encourage and gradually enlist new stakeholders by identifying and acknowledging critical external activities. This involves incorporating their modes of intervention and supported activities into the broader mission framework. This will be instrumental for the mission evaluations.

Recommendations for managing the mission portfolio

4. **Perform portfolio reviews and stage-gate portfolio mechanisms to create strategic flexibility and avoid portfolio lock-in.**

Use a stage-gated approach whenever possible to cut or change activities that are not in line with mission goals or have not produced the desired impact based on explicit and clearly communicated rules communicated from the outset to all partners. Frequent portfolio reviews and regular theory of change updates (e.g., every few years), reflecting and internalising learned lessons and changes in the broader environment, should also help avoid implementation and confirmation bias.

5. **Foster a culture of trust and experimentation and ensure risk acceptance in portfolio processes.**

Strive to establish clear goals and boundaries around experimentation so that stakeholders or mission actors feel comfortable to engage in new activities, processes, or methods. If an activity does not provide the expected desired outcome (activity failure), the mission manager should re-engage stakeholders in other activities that aim to fulfil mission objectives. In other words, the mission manager should be able to differentiate between an activity's ambition and the overall mission's ambition, emphasising that the failure of a given project and activity does not equate to mission failure. This should also be reflected in the mission monitoring and evaluation practices.

6. **Establish clear portfolio management responsibilities within the mission management team.**

As missions need to be managed at several nested levels and/or in different areas, they should also establish specific roles (like Mission Head, Mission Deputy, or Mission Lead) that oversee different levels of portfolios or sub-portfolios and facilitate the differentiation of activities and outcomes under their control. Certain missions have created a 'portfolio manager' function.

7. **Promote interlinkages and exchanges between the different projects and activities within the portfolio.**

Mission portfolio management assumes that a single-shot project or activity will not be able to deliver on the mission. Harnessing the complementarities between these projects and activities through proactive and purposive management will be beneficial to achieving the mission. It is, therefore, essential to be able to identify the different types of complementarities – and therefore needed linkages – between

projects/activities within a given mission portfolio. For example, three types of complementarities can be identified under more technology-centric missions/MOIPs: 1) activities within a given set of ‘innovation solutions’, 2) solutions at different stages of development, and 3) different potential solutions. While this still needs to be further developed, several mission managers endeavour to promote exchanges and, if possible, cooperation between various activities within their portfolio, if only under the form of regular thematic events where the different supported organisations are gathered.

Recommendations for monitoring the mission portfolio

8. Monitor the success of the mission based on the contribution of the portfolio to the realisation of the mission goal, not the aggregation of individual projects’ success

The most crucial part of mission portfolio management is the delivery of the mission rather than the number of activities or their scope. Ongoing mission portfolio monitoring should be focused on providing mission managers with the information needed to ensure that the portfolio will deliver the outcomes defined rather than the success of singular projects. This requires a pragmatic approach where selected key metrics and dimensions are tracked continuously (while avoiding administrative burden or information overload), and sense-making exercises are re-occurring to assess mission progress from different perspectives.

9. Develop specific visualisation tools to monitor mission portfolios.

Notably, portfolio visualisations should create the basis for a robust system for network analysis of the portfolio so that users and stakeholders can better understand the connection among all projects and portfolios. Some tools have been developed, such as the Future by Lund, a tool developed by Lund University for ‘Innovation Ecosystem Portfolio Tracking.’ (Commission, European, 2023^[13]). However, these remain very few and are not adapted to all mission cases. Further efforts through action-oriented experiments involving researchers and mission managers should be dedicated to developing and testing portfolio management tools in different mission contexts.

10. Integrate monitoring with learning processes to ensure continuous improvement.

Mission portfolios are not static – they must adapt based on learnings from their experimental parts and changing circumstances and the impact of established activities and projects. It is important that day-to-day portfolio monitoring is also connected to other learning processes established for the mission (mission strategic foresight activities, evaluation, sensemaking sessions, etc.). Hence, monitoring activities should feed into other learning practices and vice versa.

Next steps

Further research is needed on how mission portfolio management can, in different contexts, fulfil its core functions of sensemaking, mapping and visualisation, diversification, monitoring and evaluation, pro-active co-ordination and stewardship. More exploration is also needed on the topic of which tools are or can be made available to effectively visualise, monitor and take action with regard to mission portfolios. Yet, tools are only helpful if the individuals and roles they support are well-positioned and enabled to actively steward mission portfolios, including those that span across and between organisations and sectors. Effective MPM rests on many other practices and capacities, including actively using theories of change, developing mission-specific portfolio monitoring criteria, and embracing digital tracking tools and methods. Thus, effective portfolio management requires broader institutional shifts and development.

The Mission Action Lab will continue to monitor MPM’s emerging practice. Building on the insights provided in this paper, the Mission Action Lab aims to give practical guidance on its different aspects in the future.

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Annex A. List of key questions on mission portfolio management by building blocks

Table A.1. List of key questions on mission portfolio management by building blocks

1. Selection/design of the portfolio
• How to assure strategic alignment between the broader mission ambition, theory of change, impact pathways and initiatives the portfolios compose of? How to design portfolio aligned with the end mission goal? How should this in practice inform the selection (and prioritisation) of projects/instruments within the mission portfolio? • What are the limits or boundaries to a mission portfolio? There will likely be mission-relevant pre-existing projects in the system already, so portfolio definition can be a challenge. What should be considered in or out of the bounds of the mission portfolio? • Should missions be technology neutral? While laudable in theory, it seems that missions are often past that point (in urgency and policy-making) and instead require technology-specific deployment and scaling strategies, see for instance the goal of electrification to reach climate-neutral mobility. • How can missions ensure the directionality necessary to channel policy efforts and funding, while at the same time harnessing bottom-up innovation stemming from the passion, drive and creativity of entrepreneurial individuals in the ecosystems?
2. Steering of the portfolio
• What does good-enough mission portfolio management look like? • Need for a certain level of transparency and openness, so that all actors involved in a mission can have an overview of what is happening and how they can contribute. How to do so without overexposing the work and investment decisions, thus lowering the tolerance for risk-taking? • How to assign clear ownership over mission portfolios while keeping the portfolios open for ecosystem input?
3. Complementarity within the portfolio
• How to analyse and spot portfolio synergies between different mission instruments (projects, regulation, procurement etc.)? • How to analyse time-dependent activities in mission portfolios (stages of development, interdependencies between exploration and exploitation activities)? • To what extent support experimentation/exploration in mission portfolios versus standardisation and scaling?
4. Prioritisation of the portfolio
• How to avoid portfolio overload in situations of limited resources? How to avoid mission creep or bloat. What tactics can be used to make sure projects in the portfolio are the most efficient way to reach the mission targets? • How should mission project portfolio manager think of risk and project failure? • What balance should mission portfolios have in terms of exploration versus exploitation? The financial analogy inherent in the term portfolio suggests an inherent value of diversification. How to assure that the focus does not become incremental innovation? • How can a portfolio be mapped? What does a healthy or appropriate mission portfolio look like? What are key criteria to map and analyse within a mission portfolio? How to conduct project/programme/instrument valuation in line with

the key criteria for mission portfolios (e.g., possibility of failure and zones of acceptable risk across the portfolio)? How to avoid ambiguity of criteria to make decisions for proceeding or cutting projects?

- What is the appropriate project mix is dependent on the mission area, context and strategy?
- How are projects killed? How to give portfolio review sufficient “teeth” to cut initiatives? There is always a risk that policymakers will not cut projects that are underperforming or no longer fitting the overall mission strategy.
- When it is difficult to cull a portfolio by consensus, who has the final say? On what basis and metrics are culling decisions taken?
- How to avoid ambiguity in decision making due to lack of solid information and decision-making bias due to overconfidence in current activities in the portfolio?

5. Reach of the portfolio

- Who has the responsibility of ecosystem wide mission portfolio analysis?
- How to build ‘alliance’ portfolios which look at complimentary relationships and diversity of sources of knowledge and resources, for missions?
- How to holistically map ecosystem wide portfolios? Who should these portfolios be open to?
- How to assure both transparency and competitive interest in mission portfolios spanning a variety of organisations and strategic interests?
- How to combine broad coalition-building with clear responsibility and accountability for delivery?

6. Engagement in the portfolio

- Which methods and tools to use for collective intelligence on mission portfolios?
 - How to find time and resources for meaningful engagement on mission portfolios?
 - How to assure buy-in across various organisations for a common mission portfolio approach? Would a uniform approach be possible or even desirable?
 - How can stakeholder input be incorporated into portfolio management decisions?
 - How to assure that no one group can dominate the conclusion of the portfolio valuation?
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Annex B. Necessary tools, processes, resources and checklist questions for mission portfolio management

Table B.1. Necessary tools, processes, resources and checklist questions by MPM component

	Building the portfolio	Managing the portfolio	Monitoring the portfolio
Needed tools, processes and resources	• A mission problem definition and theory of change, with the implications in terms of activities needed to 'make it happen' (based on a systems analysis of the problem context) • An estimate of the capabilities and resource needed to achieve the theory of change • An assessment of the landscape of projects and activities relevant to the theory of change (with all necessary information on their sources of funding and ownership) • Understanding of the key portfolio activities under direct control of mission managers and stakeholders whose contribution to the (nested) portfolio is needed. Planned activities (including capability building and creating an uniform vocabulary around the mission ToC/portfolio) to get these stakeholders on board and committed to delivering towards the mission. • A clear understanding of the mission available policy mix and, more generally, of all the levers that can be mobilised to realise the missions • Visibility on available resources at least for several years ahead • A consensus on the accepted level of risks among the mission partners	• Regular portfolio sensemaking processes with key mission stakeholders to build buy-in to the mission and its portfolio • A well-accepted process to review activities and reorient/prolong/terminate them, as relevant (e.g., stage-gate process) • A competent and trusted portfolio manager (and associated team) with clearly delineated the tasks and decision power • An understanding of the relationships of cooperation and competition among the organisations involved in the portfolio • Internal or external strategic intelligence (including strategic foresight) accessible to keep track of (present and future possible) changes in the context	• An information system that allows visualising the portfolio and its evolution, in relation to the mission theory of change • Monitoring criteria allowing assessing substantive progress towards the mission realisation and tracking mission processes • Tool to collect portfolio input from contributing organisations, if possible, in real time • Reflexive mechanisms to inquire within the portfolio, draw lessons and channel resulting information to the relevant mission bodies
Check list of questions for mission portfolio managers	• What are the past and ongoing activities that do or could contribute to the mission's theory of change i) under the 'mission control sphere'; ii) outside the 'mission control sphere'? • How to mobilise and access the results of identified past and	• What similar or complementary activities should be directly connected for mutual reinforcement? • What networks/events/process/platform could maximise the flows of knowledge about challenges and	• What communication canals and reporting system are necessary to acquire sufficient and reliable information on activities' progress and challenges? • How to show progress and make justice to the collective efforts, even at an early mission stage?

	ongoing activities? • What new activities and projects would be needed to contribute to the mission realisation? How could these activities be categorised into coherent sub-groups/work packages (e.g., thematic categories, generations of solutions)? • Which of the needed activities could be supported with the policy mix available to the mission? • What modifications of the policy mix could expand the reach of the mission towards the needed activity? How to ensure necessary directionality in the mission policy mix? • Which of needed activities are out of the mission reach, and what partnerships could help mobilising/bridging to these activities? • What parts of the theory of change are unclear/uncertain and would require experimentation, exploration of various alternatives and/or linkages to more open research activities? • What would be a feasible/realistic sequencing of the portfolio given the available resources and management capacity and the expected pace of learning and progress? • What are the key activities that are essential to the mission realisation, and what is the contingency plan if these activities fail?	opportunities among the portfolio organisations? • How can relationships with potential clients/users within the portfolio be promoted from the outset? • How to enhance a sense of buy in and commitment to the mission within the portfolio, beyond each individual project? • What activities are progressing according to plans, falling behind, stagnating, showing other interesting results? • What new activities are needed? what new partners/stakeholders would benefit the mission progress? • What capabilities are needed in the portfolio management team and, if not available, how to access the necessary advice (in all matters, technical, regulatory, IP, etc.)? • How to group some activities in broader works packages where some management and orchestration tasks could be delegated to trusted stakeholders ('nested portfolio' strategy)?	• How can lessons be drawn from activities' progress and failure and effectively formalise and disseminate these lessons within the portfolio? • How can lessons be drawn from the mission in its entirety and its portfolio approach to formalise these learnings among mission partners (and beyond, if possible)?
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Endnotes

¹ See further: <https://www.undp.org/sites/g/files/zskgke326/files/2022-01/UNDP-RBAP-Sensemaking-Workshop-Preparation-Guide-public-version-2021.pdf>.

² See further: <https://www.cgiar.org/food-security-impact/new-results-dashboard>.

³ While this is notoriously difficult to track, measure, and evaluate, innovation portfolio managers should develop mechanisms to address a few key questions: 1) is the portfolio of innovation aligning activities and projects with the overall organisational purpose or mission? 2) is the portfolio creating and maintaining distinct and suitable strategies for managing different types of innovation activity? 3) are new linkages across innovation activities being made and is learning happening between them? 4) are innovation activities in the portfolio shifting based on identified gaps, changed operating environments, or new opportunities or threats?

⁴ The handbook for transformative theory of change (ToC) developed by TIPC & EIT Climate-KIC provides a methodology that helps projects, programmes and organisations to design, implement and evaluate interventions that contribute to systems transformation. The handbook also directs attention to the portfolio in which an initiative is embedded and their interactions. Overall, the ToC is proposed as a dynamic tool and will evolve as your project develops, reflecting on the learnings it gains through the process. See <https://tipconsortium.net/publication/motion-handbook-developing-a-transformative-theory-of-change>.

⁵ See e.g.: <https://ecochallenge.org/iceberg-model/>

⁶ See further: https://dashboard.tech.ec.europa.eu/qs_digit_dashboard_mt/public/sense/app/630dc6b8-23f3-43a1-a275-7a59115f3813/sheet/9758dd27-b8ac-43f3-8336-8120db0ae19a/state/analysis